Primary Protein Databases

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Introduction

- Primary protein databases are repositories that store raw data, which are primarily the results of experimental work. These databases contain sequences and basic information submitted directly by researchers and sequencing projects.
- **Features:**

→ Raw Data Storage:

Store experimentally derived protein sequences and related information. Serve as the initial point of deposition for new protein sequence data.

→ Basic Annotations:

Provide essential annotations such as protein names, sources, and sometimes basic functional information.

→ Data Submission:

Allow researchers to submit new protein sequences directly.

- **Primary Protein Dbs:**

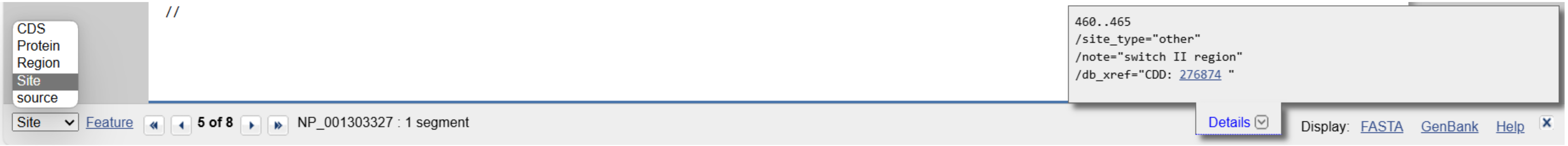
→ NCBI Protein Database (GenPept): Contains sequences from GenBank, RefSeq, and other NCBI databases.

→ PIR-PSD (Protein Information Resource - Protein Sequence Database): Contains protein sequences with basic functional annotations.

→ UniProtKB/TrEMBL: Contains protein sequences that are automatically annotated and not yet manually reviewed.

NCBI Protein Db

- ❑ GenPept: NCBI → Protein → Search for a protein of interest → Analyze
- ❑ Flatfile:
    - → Locus, Definition, accession, Organism, Reference
    - → Features:
        - source:
        - Protein
        - Region:
        - Site:
        - CDS:
    - → Origin: Amino acid sequence



PIR/Protein Information Resources

- https://proteininformationresource.org/
- The world's first database of classified and functionally annotated protein sequences that grew out of the Atlas of Protein Sequence and Structure
- Developed by Margaret O. Dayhoff - the pioneer of the field of bioinformatics - at the National Biomedical Research Foundation (NBRF) at Georgetown University, USA.
- In 1988 the NBRF joined with the MIPS, and the Japan International Protein Information Database (JIPID), to form the PIR-International
- An integrated protein informatics resource for genomic, proteomics and scientific research
- Freely accessible and provides free analysis tools
- Database is updated twice a week.
- Cross-referenced Db

PIR/Protein Information Resources

❑ The PIR maintains several databases about proteins:

→ **PIR-PSD:** the main protein sequence database

→ **iProClass:** classification of proteins according to structure and function

→ **NREF/Non-redundant reference Db:** a comprehensive non-redundant collection of over 800000 protein sequences merged from all available sources

→ **iPTMnet:** Integrated Post translational modifications.

→ **iProLINK:** Integrated protein literature information and knowledge.

→ **PIRSF:** Protein information resource, superfamily and family. Classification based on sequence similarity.

→ **UniProt:** Provides information to UniProt (secondary protein Db) along with EMBL and SIB.

PIR

- **The web server of the PIR shows the richness of information retrieval tools available:**
  - → Data classified by sequence similarity into Superfamilies, families and homology domains
  - → Retrieval of database entries, selected according to: sequence similarity, text fields in sequence or annotations, motifs and domains, structure or functional classifications, or appearance in a specified genome
  - → Statistical analysis of classification of entries
  - → Gene family classification, useful for genome annotation
  - → Launching of calculations, including pairwise or multiple sequence alignment, and sequence similarity searches
  - → Extensive links to other databases, including bibliographic databases

PIR-PSD

- Functionally annotated protein sequences
- Reference Paper:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC165487/

PIR A UniProt CONSORTIUM MEMBER
Protein Information Resource

About PIR | Resources | Search/Analysis | Download | Support

HOME / Databases / *PIR-PSD*

Database Description for PIR-PSD

Release 80.00 (31 Dec 2004) is the final release for the PIR-International Protein Sequence Database (PIR-PSD), the world's first database of classified and functionally annotated protein sequences that grew out of the Atlas of Protein Sequence and Structure (1965-1978) edited by Margaret Dayhoff. Produced and distributed by the Protein Information Resource in collaboration with MIPS (Munich Information Center for Protein Sequences) and JIPID (Japan International Protein Information Database), PIR-PSD has been the most comprehensive and expertly-curated protein sequence database in the public domain for over 20 years. In 2002, PIR joined EBI (European Bioinformatics Institute) and SIB (Swiss Institute of Bioinformatics) to form the UniProt consortium. PIR-PSD sequences and annotations have been integrated into UniProt Knowledgebase. Bi-directional cross-references between UniProt (UniProt Knowledgebase and/or UniParc) and PIR-PSD are established to allow easy tracking of former PIR-PSD entries. PIR-PSD unique sequences, reference citations, and experimentally-verified data can now be found in the relevant UniProt records.

Search PIR-PSD records with a PIR-PSD ID/Accession #:

Submit Reset

iProClass Summary Report for UniProtKB Entry: A0A1W2PQ27

ID Mapping

GENERAL INFORMATION	
Protein Name and ID	UniProtKB ID: S72L1_HUMAN; UniProtKB Accession: A0A1W2PQ27; Protein Name: RNA polymerase II subunit A C-terminal domain phosphatase SSU72 like protein 1 GenPept: BAA01853.1 ; AAC97607.1
Taxonomy	*Source Organism:* Homo sapiens (Human) *Taxon Group:* Euk/mammal *NCBI Taxon:* 9606 *Lineage:* Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.
Gene Name	SSU72L1
Keywords	hydrolase; mrna processing; nucleus; protein phosphatase; reference proteome
Function	Protein phosphatase that catalyzes the dephosphorylation of the C-terminal domain of RNA polymerase II. Plays a role in RNA processing and termination.

CROSS-REFERENCES	
Bibliography	▸ View Bibliography Information *Annotated references:* PMID: 16554811 [UniProt]
DNA Sequence	EMBL: AC018793 DDBJ: AC018793 GenBank: AC018793
Genome/Gene	*Gene Name:* SSU72L1; SSU72 like 1 *Synonyms:* SSU72P7 ; *Map Location:* 11p15.4 Entrez Gene: 390033 RefSeq: NM_001413998.1 NP_001400927.1 [Map Viewer] Ensembl: ENST00000524542.3; ENSP00000491894.1; ENSG00000284438.2
Gene Expression	CleanEx SOURCE Bgee: ENSG00000284438

FAMILY CLASSIFICATION	
UniRef	UniRef100_A0A1W2PQ27; UniRef90_A0A1W2PQD8; UniRef50_A0A1W2PQD8
Pfam Domain	Pfam: PF04722: Ssu72-like protein (6-194)
InterPro	InterPro: S72L1_HUMAN IPR036196: Phosphotyrosine protein phosphatase I superfamily IPR006811: RNA polymerase II subunit A
Other Classification	Gene3D: 3.40.50.2300 -

SEQUENCE

```
MLSSTLRVAV VCVSNVNRSM EAHSILRRKG LSVRSFGTES HVRLPGPRPN
RPVVYDFATT YKEMYNDLLR KDRERYTRNG ILHILGRNER IKPGPERFQE
CTDFFDVIFT CEESVYDTVV EDLCSREQQT FQPVHVINME IQDTLEDATL
GAFLICEICQ CLQQSDDMED NLEELLLQME EKAGKSFLHT VCFY
```

UniProt

- ❑ The Universal Protein Resource (UniProt) is a comprehensive resource for protein sequence and annotation data.
- ❑ UniProt is a collaboration between the EMBL-EBI, the SIB Swiss Institute of Bioinformatics and the Protein Information Resource (PIR).

→ EBI: Hosts large resource of bioinformatics dbs and services

→ SIB: center of Swiss-Prot. Maintains ExPASy server

→ PIR: protein sequence analysis and classification

- ❑ Across the three institutes more than 100 people are involved through different tasks such as database curation, software development and support.
- ❑ The manual annotation of an entry involves detailed analysis of the protein sequence and of the scientific literature.
- ❑ UniProt provides computational tools like BLAST, Alignment, Mapping
- ❑ The UniProt databases are the UniProt Knowledgebase, the UniProt Reference Clusters, and the UniProt Archive (UniParc).

UniProt Databases

A. **UniProt KB:**

   i. ***Swiss-Prot:***

   ii. ***TrEMBL:***

B. **Species Proteomes:**

C. **UniRef (Protein Reference Clusters):**

D. **UniParc (Sequence Archive):**

UniProt KB

A. **<u>UniProt KB/UniProt Knowledgebase:</u>**

1) Central db in UniProt and curated protein db
2) Each entry contains amino acid sequence, protein name, description, taxonomic data and citation information
3) Contains collection of functional information on proteins with accurate annotations which includes ontologies, classification and cross-reference
4) Source of data for UniProt KB: DDBJ, ENA, GenBank, Sequence of PDB structures, Sequences from RefSeq
5) Databases of UniProt KB:
   - ***i. Swiss-Prot:***
   - ***ii. TrEMBL:***

UniProt KB- SwissProt

1. An annotated protein sequence database, which was created at the Department of Medical Biochemistry of the University of Geneva and has been a collaborative effort of the Department and the European Molecular Biology Laboratory (EMBL), since 1987. The aim of UniProt KB/SwissProt is to provide all known relevant information about a particular protein.

2. SWISS-PROT is now an equal partnership between the EMBL and the SIB

3. The SWISS-PROT protein sequence database consists of sequence entries. Sequence entries are composed of different line types, each with their own format.

4. <u>Features:</u>

→ It is a manually annotated (curated by experts). The experimented and predicted data is manually verified by the experts.

→ It provides a high level of annotation such as the description of protein function, domains & sites, secondary structure, post-translational modifications, variants, etc.

→ A minimal level of redundancy. Non-redundant protein sequence database with 99% accuracy. Thus called considered as a gold standard.

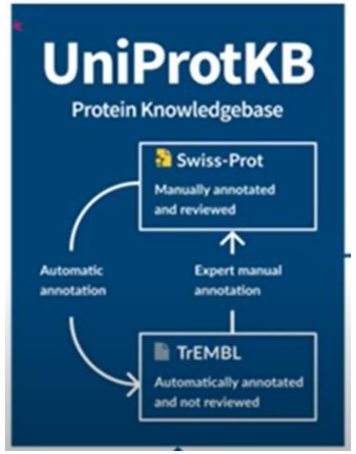
→ It combines information extracted from scientific literature and biocurator-evaluated computational analysis.

→ High level of integration with other databases

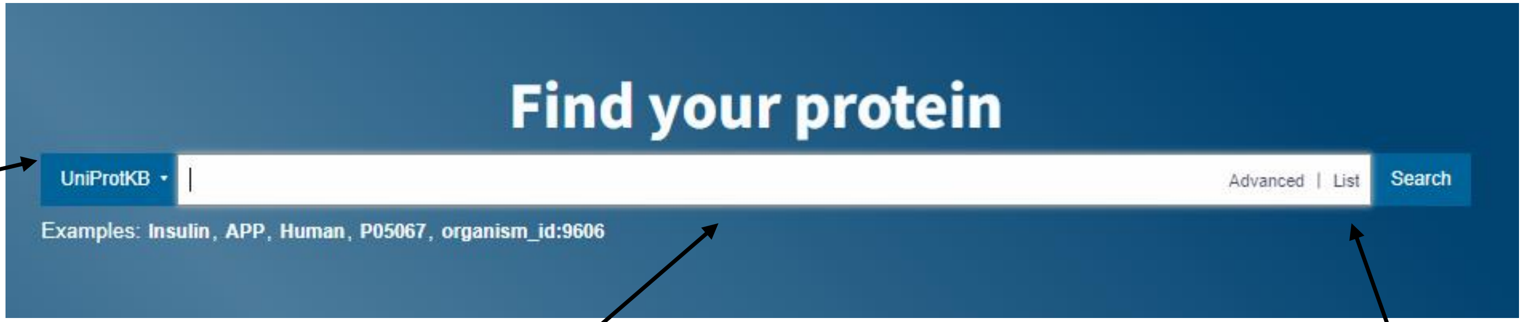
UniProt KB-TrEMBL

ii. TrEMBL:

1. Translated EMBL Nucleotide Sequence Data Library or Translation of EMBL
2. Introduced by EMBL-EBI collaborator
3. Created in 1996 in response to increased dataflow resulting from genome project
4. Contains computationally analyzed records
5. Data is automatically annotated & classified through computational tools
6. Data is unreviewed
7. This data is further manually annotated & stored into SwissProt



UniProt KB



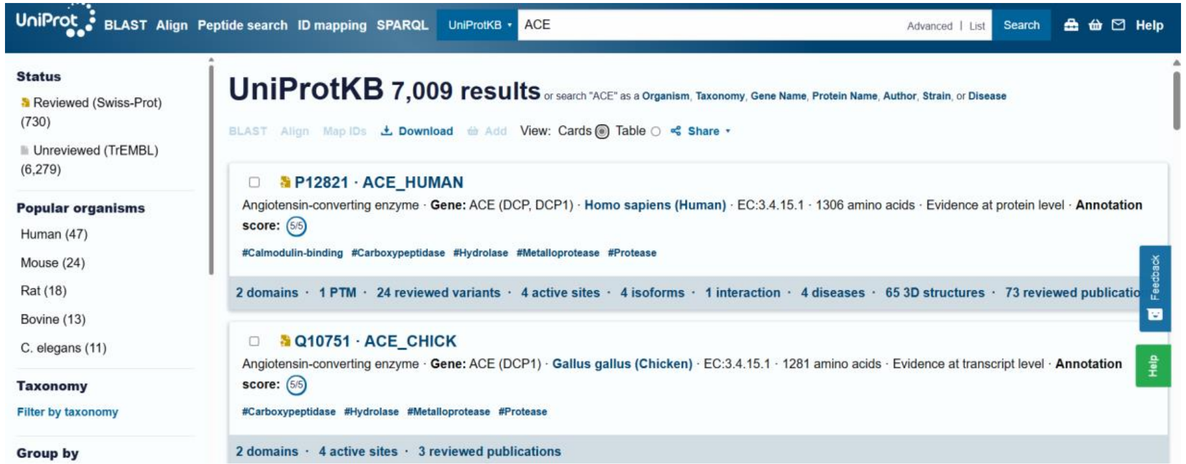
Select Db

Search gene, protein, disease

Search using accessions or IDs

UniProt KB

Reviewed or Unreviewed



Card View

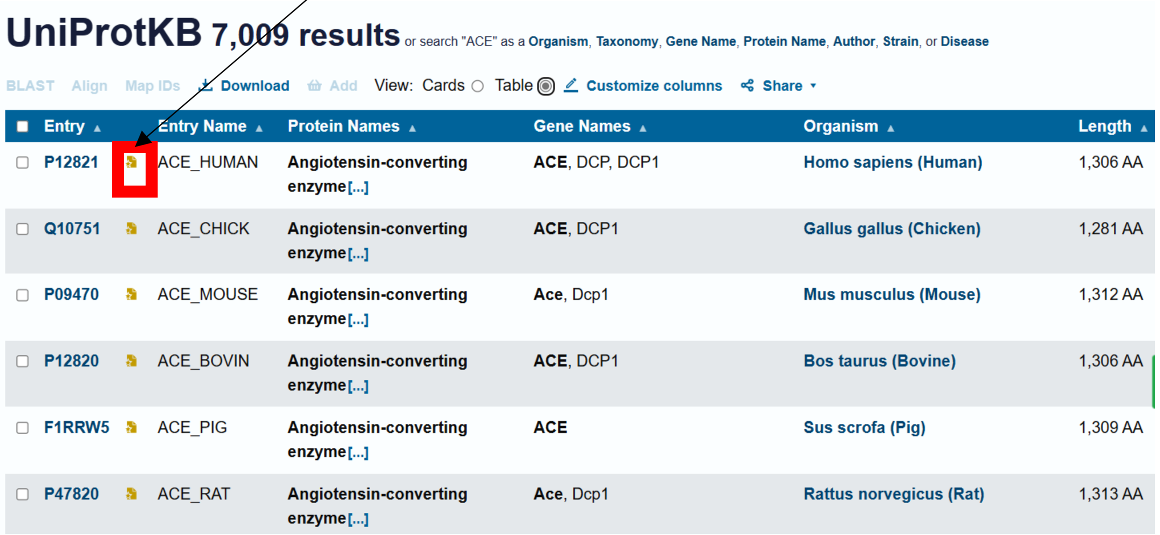
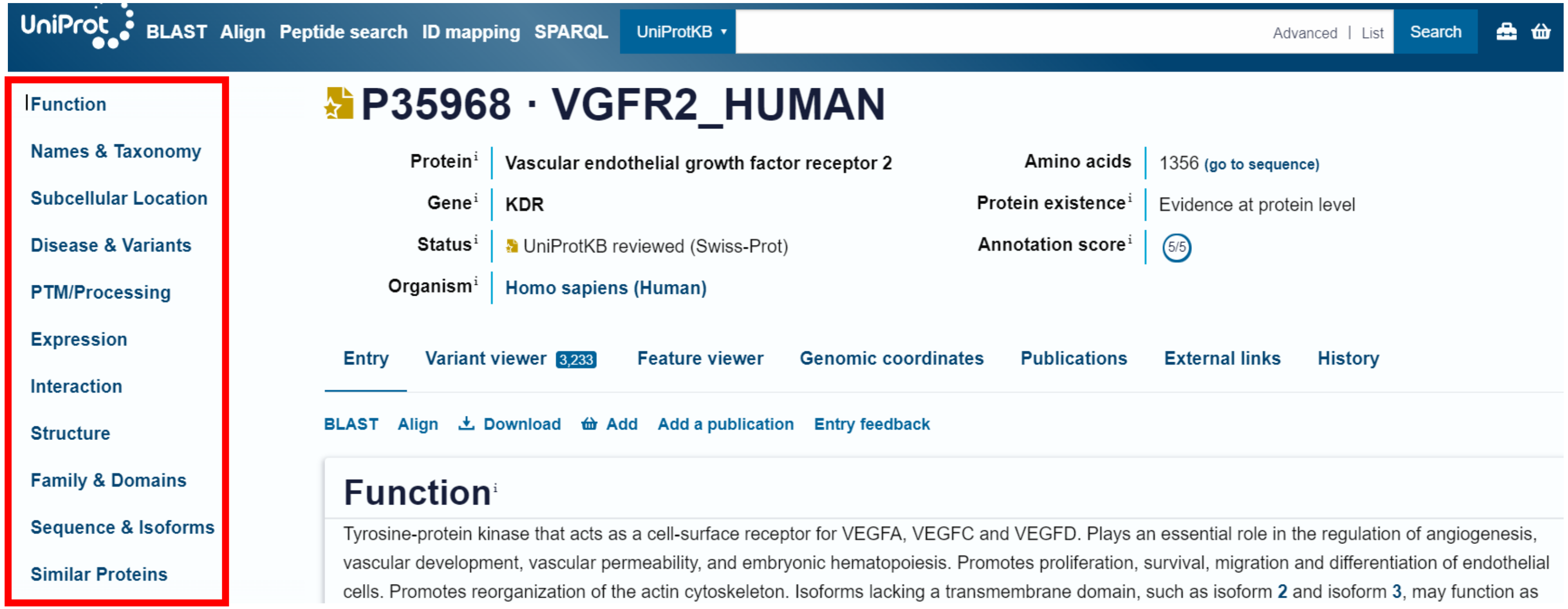
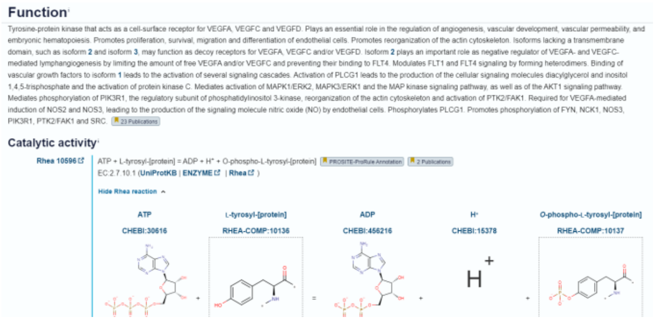


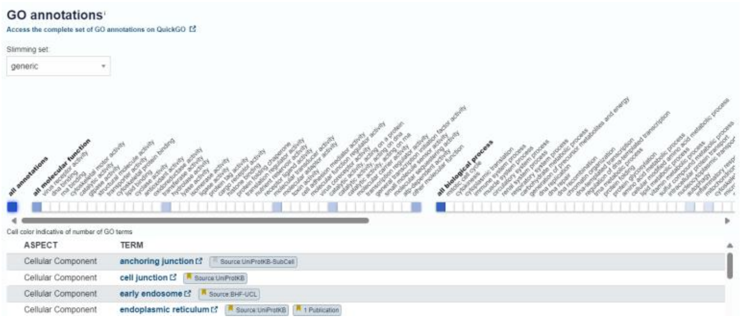
Table View

UniProt KB


UniProt
BLAST Align Peptide search ID mapping SPARQL
UniProtKB
Advanced | List
Search
Function
Names & Taxonomy
Subcellular Location
Disease & Variants
PTM/Processing
Expression
Interaction
Structure
Family & Domains
Sequence & Isoforms
Similar Proteins
P35968 · VGFR2_HUMAN
Protein: Vascular endothelial growth factor receptor 2
Gene: KDR
Status: UniProtKB reviewed (Swiss-Prot)
Organism: Homo sapiens (Human)
Amino acids: 1356 (go to sequence)
Protein existence: Evidence at protein level
Annotation score: 5/5
Entry
Variant viewer 3,233
Feature viewer
Genomic coordinates
Publications
External links
History
BLAST Align Download Add Add a publication Entry feedback
Function
Tyrosine-protein kinase that acts as a cell-surface receptor for VEGFA, VEGFC and VEGFD. Plays an essential role in the regulation of angiogenesis, vascular development, vascular permeability, and embryonic hematopoiesis. Promotes proliferation, survival, migration and differentiation of endothelial cells. Promotes reorganization of the actin cytoskeleton. Isoforms lacking a transmembrane domain, such as isoform 2 and isoform 3, may function as

UniProt KB





FlatFile manual: https://www.uniprot.org/help/uniprotkb_manual

Functions:

- General functions in detailed along with bibliographic references
- Catalytic activity: reactions catalyzed by an enzyme. Enzyme commission no. provided which is a numerical classification scheme for enzymes, based on the chemical reactions they catalyze.
- Activity regulation: regulating factors affecting the biological activity of the searched protein
- Binding site: Binding site for any chemical group (co-enzyme, prosthetic group, etc.)
- Active site: Amino acid directly involved in the activity of an enzyme
- GO annotations: Ontologies guiding through molecular, biological & cellular functions related to proteins. Cross referencing to pathway, bibliographic databases

Names & Taxonomy: protein and gene that codes for protein, source of protein

UniProt KB

Subcellular location: Location in cell, Description of the subcellular location of the mature protein (including isoform locations if available)

Diseases & Variants: diseases the protein is associated with & variants seen in that protein. Involvement in which disease, variant ID, position of variation, Natural variants of the protein that are associated with one of the diseases, mutagenesis (Site which has been experimentally altered by mutagenesis). Variation viewer, cross-referencing to other variation Dbs like OMIM, CLinGen can be explored.

PTM/Processing: Post translational modification

Expression: Description of the expression of a gene in different tissues. Description of how the expression of a gene varies according to the stage of cell, tissue or organism development

Interaction: interactions with other proteins,

Structure: secondary structure of protein & cross-referencing to protein structure databases

Family & Domains: necessary domains for activity, sequence similarities

Sequence: canonical sequence, cross-reference to nucleotide & protein sequence

Similar Proteins: 100, 90, 50% identical sequences

UniRef/UniProt Reference Cluster

1. Provides clustered set of sequences from the UniProt Knowledgebase (including isoforms) and selected UniParc records in order to obtain complete coverage of the sequence space at several resolutions.
2. The UniRef100 database combines identical sequences and sub-fragments with 11 or more residues from any organism into a single UniRef entry, displaying the sequence of a representative protein, the accession numbers of all the merged entries and links to the corresponding UniProtKB and UniParc records.
3. Composed of 3 components:

→ **UniRef100:** contains sequences which are 100% identical. Identical sequences and sub fragments are presented as single entry with accession no. of all entries.

→ **UniRef90:** contains sequences with identity of 90%

→ **UniRef50:** contains sequences with identity of 50%

UniParc

1. Capture all publicly available protein sequence data
2. Non-redundant protein sequence database
3. It handles all sequences as text stings. Sequences that are 100% identical over entire length are merged regardless of whether they are from same or different species. Names of both species is given.
4. Original data can be traced easily as their db cross reference their accession
5. Entries in UniParc are not annotated

ExPASy

- ❑ Expert Protein analysis system.
- ❑ Bioinformatics resource portal of various dbs, software and tools
- ❑ Managed by the SIB Swiss Institute of Bioinformatics
- ❑ Expasy is designed to serve researchers and professionals working in fields like proteomics, genomics, and transcriptomics
- ❑ <u>Components of Expasy:</u>
    - **Proteomics & Protein Sequence Analysis:** UniProtKb/Swiss-Prot, PeptideMass, ProtParam, ScanProsite, ClustalW (multiple sequence alignment), translation tool (converts nucleotides into amino acids/AA's)
    - **Genomics & Transcriptomics tools:** Ensembl, expression Atlas, OrthoDB
    - **Molecular interaction analysis:** STRING, IntAct
    - **Structural bioinformatics:** Swiss-model (homology modeling of proteins), SPDBV
    - **Functional genomics:** GO
    - **PTMs**
    - **Metabolic pathways:** KEGG